
Dyslexia Marker Variables

Reviving a System for Reporting Sample Characteristics in Learning Disabilities Research

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ABSTRACT

Despite warnings that the field of learning disabilities (LD) must address the issue of population heterogeneity, the LD research community still lacks operational definitions of specific learning disabilities as well as a systematic approach for reporting sample characteristics. Recently, however, a definition of dyslexia has been proposed that might signal a significant advance. This article builds on this definition of dyslexia by: (a) reintroducing the concept of marker variables, (b) proposing a strategy for developing an updated marker variable system, (c) presenting a preliminary working set of dyslexia marker variables, and (d) calling for the eventual adoption of this or similar marker variable systems to facilitate future research efforts.

tioning but who could not decode words in spite of more than adequate instruction and exposure to print.

However, with the introduction of the first federal definition (National Advisory Committee for the Handicapped, 1968) the more generic term *learning disabilities* was established as a disability category and, in the ensuing explosion of special education programs and related services, professionals in the newly formed LD field began to lump a multitude of heterogeneous disorders together in this umbrella category. Instead of continuing to document the characteristics of individuals experiencing academic difficulties case by case, emphasis shifted to setting up intervention programs targeted on a disability that was not truly understood or properly defined at the time.

Stanovich (1990), among others, accurately attributed much of the current definitional confusion in the LD field to this chain of events: "Educational practice simply 'took off' before a thorough investigation of certain foundational assumptions had been carried out" (p. 7). After the introduction of federal legislation aimed at providing services to students with LD, "the field [of learning disabilities] in some sense 'got ahead' of itself" (Stanovich, 1990, p. 7). Kavale (1995) points out that the LD field suffers from an identity crisis, finding itself "lodged between a natural science (i.e., the formal study of learning disabilities) and a human science (i.e., a helping profession)" (p. 148). After the sudden proliferation of services, the study of LD as a human science took precedence over a natural science perspective; instructional programs were implemented and services delivered, but the formal study of LD took on a secondary and lesser importance.

THE FIELD OF LEARNING DISABILITIES (LD) IS plagued by an inability to clearly identify specific learning disabilities. The statement by Kavale, Fomess, and Bender (1987) that "[a] major detour in LD continues to be definitions" (p. 5) still rings true today. One requires only a brief history of the field to gain insight as to how *generic* definitions took precedence over more *specific* operationalized definitions of LD.

Beginning with the work of Kussmaul (1877), Morgan (1896), and Hinshelwood (1900), the early history of the LD field mostly featured case study research, focused primarily on one specific type of learning disability that would come to be known as dyslexia. In particular, these scientists were among the first to carefully document word blindness in youngsters who seemed to possess average levels of cognitive func-

That is not to say that, faced with this situation, professionals failed to issue words of caution. Soon after LD became a federally recognized disability category, Keogh, Major, Reid, Gandara, and Omori (1978) warned against ignoring the natural science investigation of LD. Foremost among their concerns was that many researchers used the term "learning disabilities" in a generic sense, failing to recognize that individuals with LD comprised a heterogeneous population experiencing a multitude of learning difficulties: "Every cursory review of the literature suggests that many of the problems of definition of learning disabilities, and thus of selection and intervention, are directly attributable to the heterogeneity of individuals categorized as learning disabled" (Keogh et al., 1978, p. 18).

To address this concern, Keogh, Major-Kingsley, Omori-Gordon, and Reid (1982) proposed a system of *marker variables* to obtain a clearer picture of the individual with LD. Without the collection of participant-specific information, Keogh et al. warned, researchers would be at risk of defining their samples in different ways, gathering inconsistent data, and conducting studies that would be impossible to replicate. The marker variables proposed by Keogh et al. would be "agreed-upon reference points or 'bench marks' on which to base comparisons and generalizations of results across studies or programs, and across ages, backgrounds, or geographic locations of samples" (Keogh et al., 1982, p. 7). Once established, these markers would promote consistency in research terminology, measurement techniques, and reporting procedures.

In spite of this early warning, a clear conceptualization of LD and the development of an agreed-upon set of marker variables were not achieved. As a result, methodological flaws, such as the lack of clearly defined samples of individuals identified as having LD (e.g., Durrant, 1994; Hammill, Bryant, Brown, Dunn, & Marten, 1989), abound in the empirical investigation of LD in spite of two attempts by the Research Committee of the Council for Learning Disabilities (Rosenberg et al., 1992; Smith et al., 1984) to set minimum standards for reporting participant information.

These findings confirmed Keogh et al.'s (1978) prediction that without a clear conceptualization of LD and a marker variable system much of the research conducted during the 1980s is "largely nonreplicable and therefore presumably invalid" (Hammill et al., 1989, p. 178; see Note). Durrant (1994) indicated that the issue that Keogh et al. (1982) warned about—defining and describing heterogeneity in the LD population—is currently "almost completely ignored, despite demonstrations of its importance" (Durrant, 1994, p. 31). In short, the population of youngsters identified in the literature as having LD continues to be treated as a homogeneous population, and researchers fail to be consistent in their participant sampling and reporting procedures. A primary reason for this dilemma is the lack of a clear operational definition of learning disabilities; that is, a definition grounded in theory and considered to have construct validity based on empirical research and clinical observations.

The purpose of this article is to: (a) reintroduce the concept of marker variables originally developed and field-tested by Keogh et al. (1982); (b) propose a strategy for developing an updated marker variable system consistent with the Orton Dyslexia Society Research Committee's definition of dyslexia (cited in Lyon, 1995); (c) present a preliminary, working set of dyslexia marker variables; and (d) urge the eventual adoption of such a system in an effort to resolve the current state of definitional confusion and to facilitate future research efforts.

REINTRODUCING MARKER VARIABLES

The UCLA Marker Variable Project was carried out from 1977 to 1980. The goal of this project was to "develop a system for reporting sample characteristics in learning disabilities (LD) research" (Keogh et al., 1982, p. 3). When this system of LD markers was developed, its authors (Keogh et al., 1982) proposed that it would facilitate future research efforts by providing consistency in the sample information gathered and reported. These marker variables were touted as providing the field with an opportunity to reduce the definitional confusion present since its inception.

According to Keogh et al. (1982), LD marker variables were developed using available theoretical and empirical evidence. The project consisted of three distinct phases:

- (a) Phase 1 consisted of a comprehensive review of the research literature, the collection of a set of common sample descriptors and measures, connection of these common descriptors and measures with conceptual frameworks of LD, and the development of a set of preliminary marker variables based on this analysis;
- (b) Phase 2 consisted of a series of conferences held at UCLA where two sets of consultants were invited to refine, reformulate, add and possibly delete markers from the list collected in Phase 1; and,
- (c) Phase 3 comprised the construction of LD marker variable reporting forms that were field-tested by sending them out to 110 established LD researchers.

On completion of the three phases of the project, a marker variable guide was published along with documentation of data gathered from the project (Keogh et al., 1982).

Overall, four categories of markers were identified for use by professionals conducting research in the LD field:

- (a) *Descriptive markers*, used primarily to provide basic demographic and background information about participants;

- (b) *Substantive markers*, used to provide information directly applicable to individuals with LD;
- (c) *Topical markers*, designed to provide more specific and complete participant information; and,
- (d) *Background markers*, used to provide context or background for the interpretation of findings and to identify control group data.

Keogh et al. (1982) urged researchers in the field to adopt this system. They offered it as one method to deal with sample variability problems facing investigators studying LD. Unfortunately, this marker variable system is not currently being used by researchers (Durrant, 1994). One reason for LD researchers' failure to employ marker variables might be the continued definitional confusion. Although they developed a set of marker variables to describe individuals with LD, Keogh et al. (1982) stated that "it is our view that there is *no single learning disability* [emphasis added], but rather a set of at least partially independent conditions subsumed under the broad rubric of learning disabilities" (p. 72). In an attempt to address this definitional confusion, educators have recently begun to propose operational definitions for specific learning disabilities such as dyslexia.

DEVELOPING AN UPDATED MARKER VARIABLE SYSTEM

Echoing the remarks of Keogh et al. (1982), Stanovich (cited in Lyon, 1995) suggested that the LD field's attempt to define generic LD comprehensively was a large part of the problem: "scientific investigations of some generically defined entity called 'learning disability' simply make little sense given what we already know about heterogeneity across various learning domains" (p. 5). Instead of examining LD in this fashion and ignoring that heterogeneity exists, future research should "define groups specifically in terms of the domain of deficit (reading disability, arithmetic disability)" (Stanovich, cited in Lyon, 1995, p. 5). In other words, different definitions should be formulated for a variety of learning disabilities, each one pertaining to a specific deficit.

Although previous attempts had been made to provide more refined global or generic definitions of LD (National Joint Committee on Learning Disabilities, 1994; see also Hammill, Leigh, McNutt, & Larsen, 1981), the Orton Dyslexia Society Research Committee (ODSRC) was among the first organizations in the LD field to propose a domain-specific definition of one type of learning disability—dyslexia:

Dyslexia is one of several distinct learning disabilities. It is a specific language-based disorder of constitutional origin characterized by difficulties in single word decoding, usually

reflecting insufficient phonological processing. These difficulties in single word decoding are often unexpected in relation to age and other cognitive and academic abilities; they are not the result of generalized development disability or sensory impairment. Dyslexia is manifest by variable difficulty with different forms of language, often including, in addition to problems with reading, a conspicuous problem with acquiring proficiency in writing and spelling, (cited in Lyon, 1995, p. 9)

Using the ODSRC definition of dyslexia as a starting point, Lyon (1995) proposed five criteria to evaluate this and future definitions of domain-specific learning disabilities. The domain-specific LD definition must be: (a) "theory driven"; (b) "supported by a substantial body of convergent research and clinical information"; (c) based on research with "well described samples of subjects"; (d) based on theoretical constructs that are "internally valid, and that can be measured objectively"; and (e) "externally valid and useful" (Lyon, 1995, pp. 7-8).

To verify that the ODSRC definition satisfied these criteria, Lyon (1995) provided a rationale for inclusion of specific definitional statements organized according to the following categories:

- (a) Definitional Context, associated with the statement "dyslexia is one of several distinct learning disabilities";
- (b) Basic Definition, associated with the statement "It is a specific language-based disorder of constitutional origin characterized by difficulties in single word decoding, usually reflecting insufficient phonological processing abilities";
- (c) Unexpected Underachievement, associated with the statement "These difficulties in single word decoding are often unexpected in relation to age and other cognitive and academic abilities";
- (d) Exclusionary Criteria, associated with the statement "They are not the result of generalized development disability or sensory impairment"; and
- (e) Functional Impact, associated with the statement "Dyslexia is manifest by variable difficulty with different forms of language, often including, in addition to problems with reading, a conspicuous problem with acquiring proficiency in writing and spelling."

These categories proved consistent with both the federal (National Joint Committee on Learning Disabilities, 1988, p. i) and the National Joint Committee on Learning Disabilities

ties' (1994) definitions of LD. They also provided an essential link between domain-specific definitions of learning disabilities and marker variable systems.

Lyon (1995) attempted to justify the inclusion of these specific statements in the ODSRC definition of dyslexia by citing current theoretical constructs and empirical research. In the Basic Definition category, for example, Lyon cited the work of several researchers (Adams, 1990; Adams & Bruck, 1995; Brady & Shankweiler, 1991; Goswami & Bryant, 1990; Gough, Ehri, & Treiman, 1992; Gough & Tunmer, 1986; Liberman, Shankweiler, & Liberman, 1989; Perfetti, 1985; Share & Stanovich, 1995) that supported the statement that "[Dyslexia] is a specific language-based disorder ... usually reflecting insufficient phonological processing abilities" (Lyon, 1995, p. 9). Lyon provided similar theoretical and empirical evidence for each of the five categories, thereby satisfying the first two criteria—that definitions must be theory driven and supported by convergent research and clinical information—by which he asserted that all domain-specific definitions of learning disabilities should be judged.

Closer inspection of the five criteria proposed by Lyon (1995) shows that not two but three of his proposed criteria were satisfied: (a) the ODSRC definition was certainly theory driven; (b) each component was adequately supported by existing empirical research; and, given his thorough review of the literature, (c) the definition was definitely based on theoretical conceptualizations of dyslexia. However, two of the five criteria cited by Lyon have not yet been satisfied; in particular, without a system of marker variables to address sample variability, a definition of dyslexia cannot be based on well-defined subject samples. Without such a marker variable system, and without clearly defined subject samples, it would be impossible to ascertain whether or not the proposed definition is externally valid and useful. In other words, a truly operational definition of dyslexia cannot be formulated without a set of established criteria or marker variables.

To satisfy this definitional criterion, a strategy for developing a set of dyslexia marker variables is proposed here. This marker variable system must be derived from domain-specific definitions of LD. Generic conceptualizations of LD have proven more of a hindrance than a help to guide research efforts. The choice of dyslexia as the prototype for domain-specific definitions is a logical one, given the multitude of studies already conducted nationwide. Moreover, this marker variable system must be developed concurrently with its associated domain-specific definition. Furthermore, both the domain-specific definition and the marker variable system must conform to an accepted set of criteria, such as those proposed by Lyon (1995): They must be theory driven, supported by convergent research, grounded on internally valid theoretical constructs subject to objective measurement, based on well-defined subject samples, and ultimately proven to be externally valid. As the ODSRC has already proposed a definition of dyslexia, the first step has been taken. The next step involves the development of a working dyslexia marker variable system to better define subject samples.

A WORKING SET OF DYSLEXIA MARKER VARIABLES

Anticipating that the original set of LD marker variables would evolve as it was put to use in the field, Keogh et al. (1982) recognized their work as a first attempt to guide researchers in the description of individuals with LD. In this same spirit, an updated working set of marker variables, targeted to domain-specific definitions of LD, should be developed to allow for the field to improve its sample-reporting practices.

Before describing this new set of dyslexia marker variables, I must emphasize that this is a preliminary *working* set. In other words, this proposal should be viewed as similar to Phase 1 of Keogh et al.'s (1982) original project. A preliminary set of variables is developed based on the ODSRC definition of dyslexia. This preliminary set will need to be refined and reformulated by experts in the field, and eventually field-tested by sending a reporting form out to a representative sample of researchers investigating dyslexia.

In this the first phase of the process, each of the marker variable categories identified in the UCLA Marker Variable Project is adopted, after which individual markers are either deleted or added to achieve consistency with the ODSRC's definition and Lyon's (1995) statement categories (i.e., Definitional Context, Basic Definition, Unexpected Underachievement, Exclusionary Criteria, and Functional Impact). A rationale is provided for all deleted and added markers, with *added marker variables* appearing in italics. The marker variables presented in Table 1 are referred to as dyslexia marker variables, as the entire set, including old and new markers, would be used when describing participant samples identified as having dyslexia.

Dyslexia Descriptive Markers

These markers were designed by Keogh et al. (1982) to capture information relevant to all research conducted with human participants, including demographics, language, education, and health. Because these markers are applicable to *all* research involving human participants, all eleven of the original UCLA marker variables (see Table 1) have been retained without changes.

Dyslexia Substantive Markers

These markers were described by Keogh et al. (1982) as closely associated with definitions of LD. As originally proposed, the variables included data on intellectual ability, educational achievement, and behavioral and emotional adjustment. As the set of dyslexia marker variables is designed to be consistent with the ODSRC operational definition of dyslexia, I recommend several substantial changes. The original markers proposed by Keogh et al. (1982) were derived from "most definitions of learning disabilities" (p. 84) and included general intellectual ability, reading achievement, arithmetic

TABLE 1. Dyslexia Marker Variable Guide

Marker category	Abbreviation	Marker variable
Descriptive markers	DDM1	Sample size by sex
	DDM2	Chronological age
	DDM3	Grade level(s)
	DDM4	Locale (rural, suburban, urban, etc.)
	DDM5	Race/ethnicity
	DDM6	Sample source (public school, private school, residential facility, etc.)
	DDM7	Socioeconomic data
	DDM8	Language
	DDM9	Educational history
	DDM10	Educational placement
	DDM11	Physical and health status
Substantive markers	DSM1	<i>Language functioning</i>
	DSM2	<i>Phonological processing abilities</i>
	DSM3	<i>Single-word decoding skills</i>
Topical markers	DTM1	<i>Constitutional origin</i>
	DTM2	<i>Cognitive functioning</i>
	DTM3	<i>Reading achievement</i> (skills other than phonological processing abilities and single-word decoding)
	DTM4	<i>Proficiency in written language</i>
	DTM5	<i>Proficiency in spelling</i>
Background markers	DBM1	Year of study
	DBM2	Geographic location
	DBM3	Exclusionary criteria
	DBM4	Control/comparison group

achievement, and behavioral and emotional adjustment. Several of this original set of markers remain pertinent and consistent with the ODSRC operational definition of dyslexia.

Because I propose a separate system of marker variables specific to dyslexia, the category of Definitional Context identified by Lyon (1995) has already been addressed. The second category, Basic Definition, contains four definitional elements that require an associated set of substantive marker variables: (a) dyslexia is a *language-based disorder*, (b) usually reflecting insufficient *phonological processing* abilities, (c) characterized by difficulties in *single-word decoding*, and (d) of *constitutional* origin. As each of these elements are contained in the ODSRC definition, they must also be represented by dyslexia substantive marker variables. Information should be gathered that at least measures language functioning, phonological processing abilities, and single-word decoding skills. There is strong empirical support for inclusion of each of these elements as substantive markers: language-based disorder (Adams, 1990; Adams & Bruck, 1995; Share & Stanovich, 1995), insufficient phonological processing (Brady & Shankweiler, 1991; Goswami & Bryant, 1990; Stanovich, 1993), and difficulties in single-word decoding (Beck & Juel, 1995; Olson, Forsberg, Wise, & Rack, 1994; Stanovich, 1993). The fourth element—constitutional origin—although supported by increasing empirical evidence for inclusion in the substantive marker variables (Galaburda,

1991; Hynd, Semrud-Clikeman, & Lyytinen, 1991; Pennington, 1995; Shaywitz, Fletcher, & Shaywitz, 1995), remains a tentative aspect of the ODSRC definition according to Lyon (1995):

Due to the tentative nature of the findings derived from the genetic and neurobiological studies of dyslexia, a note of caution is extended to the researcher and the clinician. Until the next generation of studies addressing the constitutionality of dyslexia are published, the inclusion of this statement within this definition reflects a working hypothesis, (p. 14)

For this reason, the last element of Lyon's Basic Definition statement category has been excluded from the suggested preliminary list of dyslexia substantive markers.

Although general intellectual functioning was originally contained in the UCLA Marker Variable Guide as a substantive marker, I decided not to include it in the dyslexia substantive marker section of the dyslexia marker variable guide (see Table 1) because the LD field does not support the use of discrepancy-based definitions of dyslexia (e.g., Fletcher et al., 1994). However, because of the controversial nature of this proposal, and because this variable still appears in federal and state definitions of learning disabilities (Frankenberger &

Fronzaglio, 1991), cognitive functioning was retained in the dyslexia topical marker category.

Other substantive markers from the UCLA marker variable guide that were not retained in this section are reading achievement, arithmetic achievement, and behavioral and emotional adjustment. Because the proposed set of dyslexia marker variables deals with learning disabilities specific to the domain of reading, arithmetic achievement need not be represented; instead, a marker variable system specific to the domain of mathematics (i.e., dyscalculia marker variables) could be developed in a fashion similar to the system proposed here. Behavioral and emotional markers were excluded from this section because they were not consistent with the ODSRC definition.

Thus the following dyslexia substantive markers are proposed: (DSM1) *language functioning*, (DSM2) *phonological processing abilities*, and (DSM3) *single-word decoding skills* (see Table 1). Each of these markers is both consistent with the ODSRC operational definition of dyslexia and supported by a substantial body of convergent empirical research.

Dyslexia Topical Markers

As defined by Keogh et al. (1982), topical markers referred to "particular content areas or topics often associated with learning disabilities" (p. 59). These markers were characterized as identifying specific content areas that, in some cases, might be either the central focus of research or a criterion to delineate subgroups. Of all of the marker categories that Keogh et al. proposed, this was the most tentative. Compared with substantive markers, which could be measured in certain instances using standardized instruments, topical markers were not as clearly defined or as easily measured. For example, the topical marker *motivation* was offered as a variable that might have different operational definitions depending upon the investigator conducting the research. Keogh et al. characterized topical markers as tentative, different from one study to the next, and dependent on the research issue of interest. Moreover, Keogh et al. emphasized that each topical marker variable would have to be defined operationally, and researchers were urged to provide a description of specific techniques used to measure the variable of interest.

In the original UCLA marker variable guide, a total of eight topical markers were identified: (a) activity level (hyper- or hypoactivity), (b) attention (selective or sustained), (c) auditory perception, (d) fine motor coordination, (e) gross motor coordination, (f) memory, (g) oral language, and (h) visual perception.

Using Lyon's (1995) definitional components as a guide, the Basic Definition statement "dyslexia is ... of constitutional origin" (p. 9) is included in this section. Although this statement was not associated with a dyslexia substantive marker due to the "tentative nature of findings" (Lyon, 1995, p. 14), I deemed it appropriate to include it as a dyslexia topical marker. This decision was prompted by the increasing quantity of empirical research investigating the relationship

between genotypes and inherited forms of dyslexia (Cardon et al., 1994; Pennington, 1995), neuroanatomical structures and reading disabilities (Galaburda, 1991; Hynd et al., 1991), and regional blood flow and difficulties in performing phonological awareness tasks (Wood, 1990; Wood, Felton, Flowers, & Naylor, 1991).

Lyon's (1995) definitional statement category of Unexpected Underachievement was also used in the selection of dyslexia topical markers. Given the prevalence of the severe discrepancy criterion in the federal and most state definitions of LD (Mercer, Hughes, & Mercer, 1990), this element cannot be disregarded. However, due to the controversial nature of this element (e.g., Stanovich, 1991), I decided not to include it in the dyslexia substantive markers section. Instead, as with the constitutional origin element, information pertaining to cognitive functioning and reading achievement (i.e., skills other than phonological processing abilities and single-word decoding, which were included in the substantive marker variables) was designated for the topical marker variables section. Although discrepancy-based definitional criteria have been criticized as a means of identifying individuals with dyslexia (Fletcher et al., 1994; Shaywitz, Fletcher, Holahan, & Shaywitz, 1992), some researchers still propose that dyslexia is best characterized as a localized deficit not necessarily associated with broader deficiencies in cognitive functioning (e.g., Stanovich, 1992).

Finally, the definitional statement "dyslexia is manifest by variable difficulty with different forms of language, often including, in addition to problems in reading, a conspicuous problem with acquiring proficiency in writing and spelling" (Lyon, 1995, p. 9), subsumed under Lyon's (1995) Functional Impact category, has been used to include two additional dyslexia topical markers: (a) proficiency in written language, and (b) proficiency in spelling. A body of research has shown that phonological processing deficits often co-occur with deficiencies in written language and spelling in individuals with dyslexia (Fowler & Liberman, 1995; Lindamood, 1994; Moats, 1994).

Given these identified elements directly associated with the ODSRC definition, I propose the following dyslexia topical markers: (DTM1) *Constitutional Origin*, (DTM2) *Cognitive Functioning*, (DTM3) *Reading Achievement*, (DTM4) *Proficiency in Written Language*, and (DTM5) *Proficiency in Spelling*. Each of these proposed dyslexia topical markers is consistent with the ODSRC definition of dyslexia and reflects current research investigating the heritability of dyslexia, unexplained underachievement in individuals with dyslexia, and the co-occurrence of reading deficits with deficiencies in written expression and spelling.

Because this preliminary set of dyslexia marker variables includes only elements directly connected with the ODSRC definition of dyslexia, all the original LD topical marker variables developed by Keogh et al. (1982) have been deleted from this section. This does not imply that, for example, the LD marker variable of activity level is unworthy of further study. It is even possible that certain topical markers

might be moved into the substantive marker category if supported by stronger empirical evidence. Furthermore, given the rapidly expanding subtype database (Hooper & Willis, 1989), it is likely that different dyslexia topical markers will be developed by different research teams depending on their area of investigation. For example, Pirozzolo (1979) essentially employed a system of topical marker variables when he investigated performance differences between two distinct dyslexia subtype groups, one group having distinct auditory deficits, and the second group having specific visual-spatial deficits. Members of the visual-spatial subtype were identified using the three proposed dyslexia substantive markers (i.e., language functioning, phonological processing abilities, and single-word decoding), one proposed dyslexia topical marker (cognitive functioning), and six possible but not yet proposed dyslexia topical markers (right-left orientation, finger agnosia, early evidence of mirror or inverted writing, spatial dysgraphia, letter and word reversals in spelling, and faulty eye movements).

I must emphasize that dyslexia topical markers can be added to or removed from the list presented here depending on the research question at hand. As Keogh et al. (1982) stated, "it is unlikely that a single investigator will have data on all topical markers" (p. 60). However, no matter which dyslexia topical marker is used, an operational definition must be provided and measurement techniques must be described in detail.

Using Pirozzolo's (1979) subtyping research as an example, each of the possible topical markers listed for identification of members of the visual-spatial subtype would require a specific operational definition as well as a detailed description of how measurement was achieved. The finger agnosia variable, for example, would have to meet both of these requirements to be properly included as a topical marker.

Dyslexia Background Markers

The final category of marker variables developed by Keogh et al. (1982), background marker variables, was used to clarify sample comparability by providing information on year of study (i.e., beginning and ending dates of data collection), geographic location, exclusionary criteria (i.e., characteristics used to exclude subjects at the sample selection stage), and control/comparison group (i.e., whether a control group was used). According to Keogh et al., one of these background marker variables, exclusionary criteria, might contain information also located in the substantive and topical categories.

The exclusionary criteria marker variable is consistent with Lyon's (1995) statement category of the same name. The ODSRC definition of dyslexia states: "They [difficulties in single-word decoding] are not the result of generalized development disability or sensory impairment" (Lyon, 1995, p. 9). That is, dyslexia is not caused by mental retardation or hearing or visual impairment. Each of these elements can be considered exclusionary criteria. Exclusionary criteria have

therefore been retained in the proposed preliminary set of dyslexia background marker variables (see Table 1).

Keogh et al.'s (1982) other three background marker variables have been retained because they represent standard information reported in empirical research (Campbell & Stanley, 1963).

DISCUSSION

The domain-specific definition of learning disabilities, represented by the ODSRC definition of dyslexia, as well as the development of a corresponding set of dyslexia marker variables, contributes to the participant reporting procedures proposed by the Research Committee of the Council for Learning Disabilities (Rosenberg et al., 1992) and directly addresses the problem of sample heterogeneity that has been largely ignored by the LD research community (Durrant, 1994).

The definition of dyslexia proposed by the Orton Dyslexia Society Research Committee (cited in Lyon, 1995) was used to develop a preliminary set of 23 dyslexia marker variables. This tentative set of dyslexia descriptive markers (DDM), dyslexia substantive markers (DSM), dyslexia topical markers (DTM), and dyslexia background markers (DBM) is shown in Table 1. These proposed dyslexia marker variables are consistent with the seminal work on LD marker variables of Keogh and her colleagues at UCLA (Keogh et al., 1982).

Because this set of dyslexia marker variables is offered as a starting point, these markers are only tentative at best. As Keogh et al. (1982) emphasized, modifications and improvements to LD research practices must come from within the LD field, not from external sources. I propose this set of dyslexia marker variables in the spirit of improving the way in which professionals describe samples of individuals with LD. It is not sufficient to simply describe samples as generically representative of individuals with LD. Instead, domain-specific types of learning disabilities need to be defined, followed by the development of preliminary marker variables associated with each domain-specific definition. Once this process is under way, the LD field can refine these domain-specific definitions, making them truly operational with agreed-upon criteria.

The proposed set of dyslexia marker variables, based on current theoretical conceptualizations of reading and supported by a substantial body of convergent research and clinical observations of dyslexia, must now undergo the same process used by Keogh et al. (1982) to develop their UCLA marker variable guide: A second phase should be conducted, involving consultation with research investigators to determine the feasibility and utility of the dyslexia marker variables, as well as to refine this preliminary set. A third phase would involve field-testing a refined set of dyslexia marker variables, culminating in an updated and revised dyslexia marker variable guide. I predict that such a guide would have considerable value to researchers involved in the investiga-

tion of dyslexia. If this endeavor proves successful, a similar process might be used for other domain-specific learning disabilities such as dysgraphia and dyscalculia.

Until this important work is undertaken, the LD field will continue to find itself unable to rely on an accumulated body of research that is valid and replicable. The LD field must progress from its traditional adherence to a vaguely defined, generic description of LD to more domain-specific definitions such as the ODSRC definition of dyslexia.

While moving toward more domain-specific definitions of LD, however, the LD field must avoid the temptation of simply readopting a typological mentality, describing dyslexia and other specific learning disabilities as single "disease entit[ies]" (Ellis, 1985, p. 172). Instead, researchers should adopt what Ellis describes as the "notion of degrees of dyslexia" (p. 172). The investigation of LD subtypes (Hooper & Willis, 1989) has provided the LD field with a framework for delineating heterogeneous groups of LD while maintaining that each of these disorders occurs along a continuum. It is no longer adequate to characterize individuals as simply having dyslexia; it is important to describe the severity of reading problems in these individuals compared to others with dyslexia as well as to other poor readers. Researchers should ensure that each type of learning disability be contextualized within the parameters of its specific domain.

Once the LD field addresses the issue of sample variability in a systematic fashion—through the development and adoption of domain-specific definitions and sets of associated marker variables—the important work of determining the external validity of these definitions and marker variables with respect to differential treatment and education interventions can begin. ■

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NOTE

In a study conducted by Hammill et al. (1989), 277 articles published in 10 major journals were evaluated using seven criteria for describing a study's sample proposed by the Research Committee of the Council for Learning Disabilities (Smith et al., 1984). Of these 277 articles, covering studies taking place from 1984 to 1987, only 4 met the standard or satisfied all seven criteria.

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